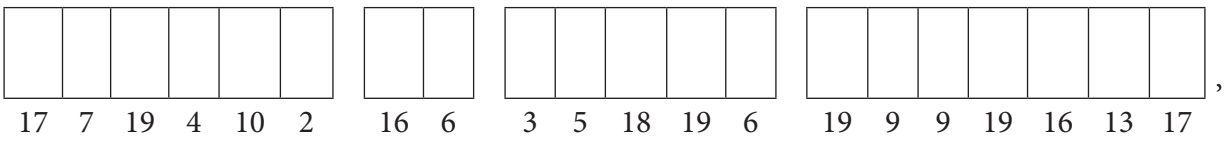
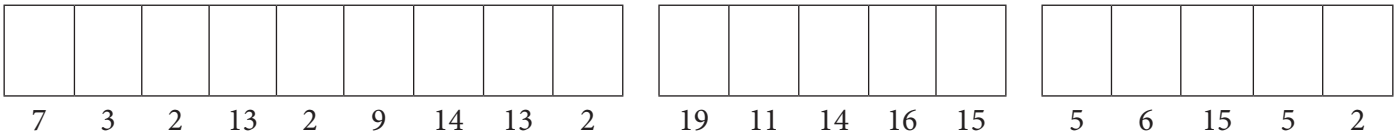
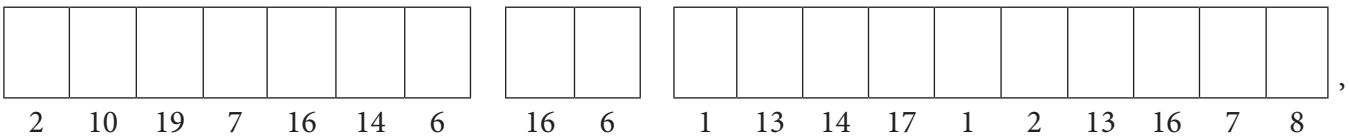
WORKSHEET

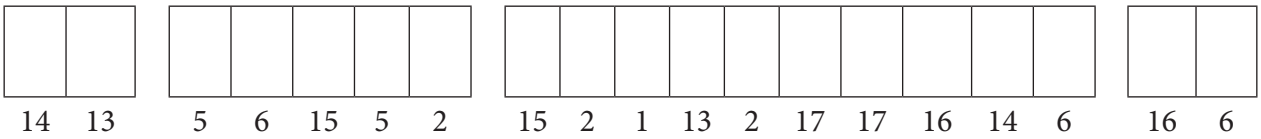
Chapter 8: The standard normal curve

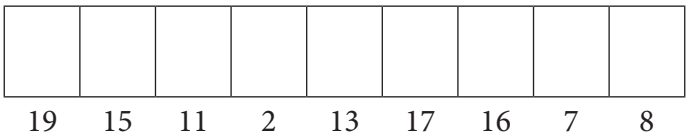
Remember that there is nothing …











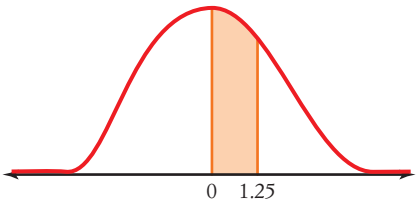
Socrates – Ancient Greek philosopher

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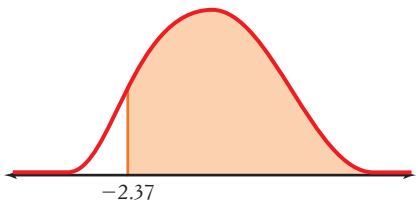
Questions

1 Find the probability of the standard normal variable for each of the following shaded areas.

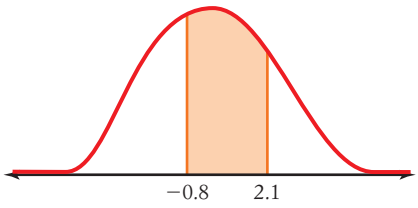
a



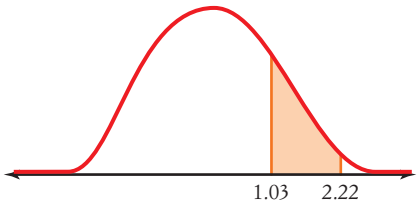
c



b



d



2 For a standard normal variable, z, find the following probabilities.

a $P(0 \leq z \leq 0.75) =$

b $P(-1.23 \leq z \leq 0) =$

c $P(-0.4 \leq z \leq 0.87) =$

d $P(z \geq -2.1) =$

e $P(1.4 \leq z \leq 2.04) =$

Solutions

A 0.54

B 0.1383

D 1.21

E 0.7702

F 0.0601

G 1.8

H 0.9911

I −0.2

L −1.49

M 2.93

N 0.3907

O 1.45

P 0.3944

R −1.63

S 2.01

T 0.4632

U 0.2734

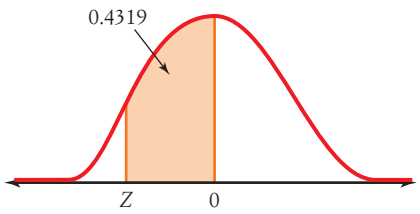
V 1

Y 0.9821

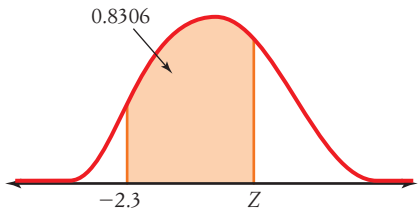
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3 Given the following probabilities of shaded areas, find the standard normal scores, z.

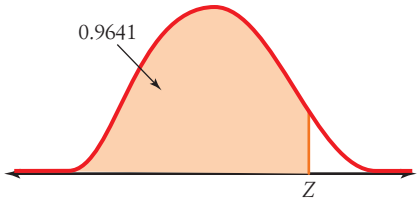
a



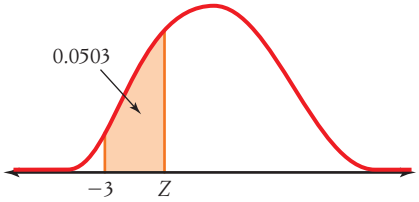
b



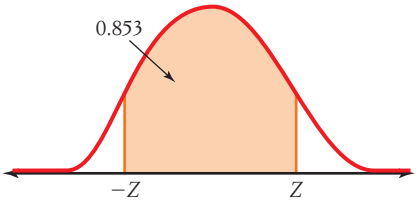
c



d



e



4 Find the standard normal score, x, for each of the following probabilities.

a $P(0 \leq z \leq x) = 0.3869$

b $P(x \leq z \leq 0) = 0.0793$

c $P(z \geq x) = 0.0222$

d $P(z \leq x) = 0.9983$

e $P(-x \leq z \leq x) = 0.4108$

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